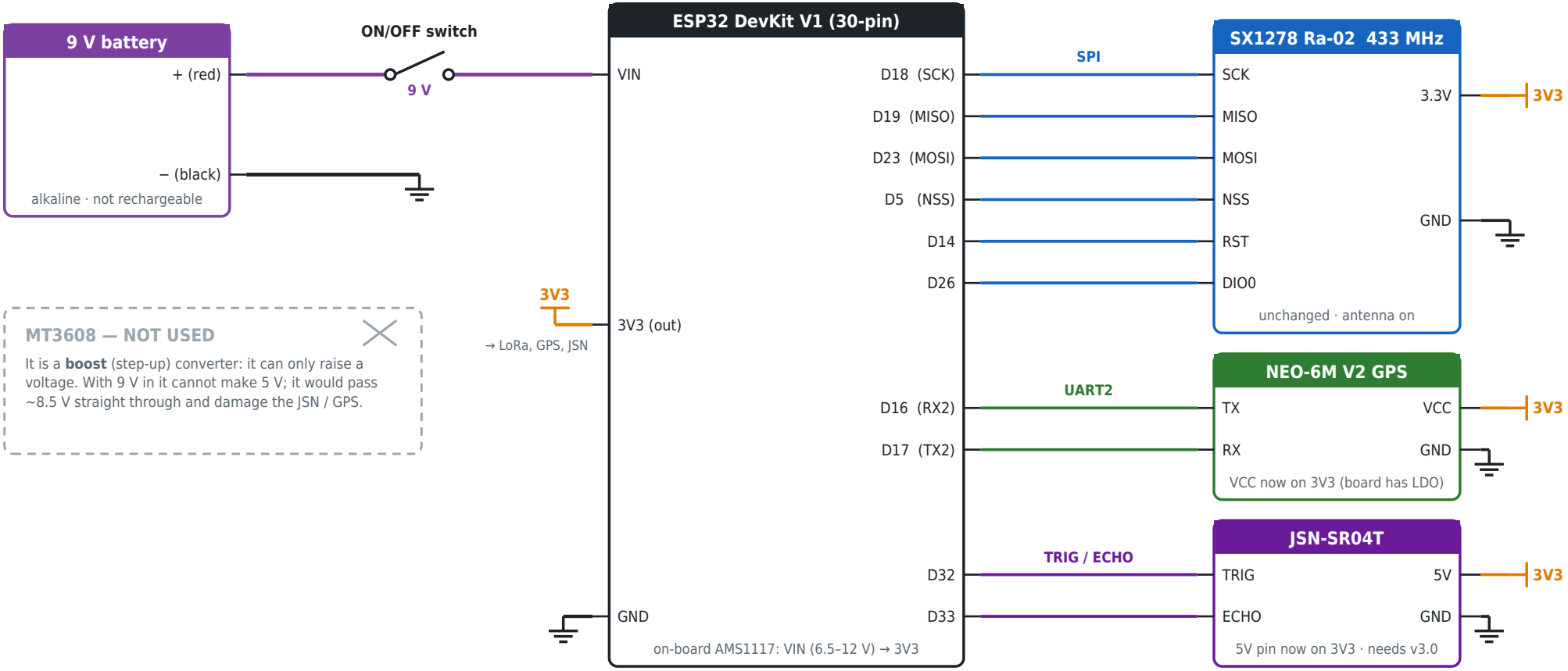
Flood Sensor — Temporary 9 V Battery Power (Rev 1b)

Replaces the LiPo + TP4056 + MT3608 power section until a new LiPo arrives. All signal wiring is unchanged.

Supplement to Flood_Sensor_Wiring_Guide.pdf
25 Oct 2026



Wiring changes vs. the guide

- 9 V **+** → switch → ESP32 **VIN**; 9 V **−** → ESP32 **GND**.
- TP4056, LiPo and MT3608 removed.
- GPS VCC and JSN 5V pin move from the +5V rail to **3V3** (there is no 5 V on the board when powered from VIN).
- Every signal wire (LoRa, GPS TX/RX, TRIG/ECHO) stays exactly as before. No code change.

Before switching on

1. **Check polarity twice**: red = +, black = −. Reversed 9 V destroys the ESP32 regulator.
2. Switch OFF while uploading code over USB.
3. Antenna on the Ra-02.
4. JSN-SR04T on 3.3 V only works on the **v3.0** board (3–5.5 V). If it shows “No echo” on 3V3, you need a 5 V source (box on the right). 3.3 V will not damage it.

Limits of this temporary setup

- Runtime: a 9 V alkaline holds ~400–550 mAh; this circuit draws ~120–150 mA → about **3–4 hours**.
- The ESP32’s small regulator drops 9 V → 3.3 V and gets **hot**. Random resets = too hot or battery flat.
- Better 5 V options: a USB **power bank** into the ESP32 USB port (then use the original 5 V wiring), or a 7805 / buck (step-down) module.

Parts: ESP32 DevKit V1 · SX1278 Ra-02 433 MHz · NEO-6M V2 GPS · JSN-SR04T · 9 V battery + snap · SPST switch

Match pins by the silkscreen label printed on each board (D14 = GPIO14).